CENTER DISPLAY [WITH CENTER DISPLAY]

id0920zz113600

Purpose

- The center display displays entertainment system and vehicle information.

Function

- The center display displays video images based on the signal sent by the LVDS*1 signal from the connectivity master unit (CMU).
- The center display is touch-type LCD. The display touch position information is sent by LVDS*1 signal to the CMU.
- The LVDS communication pathway between the CMU and the center display corresponds to HDCP*2.

*1 : Low-Voltage Differential Signaling indicates the electrical characteristics of the signal. LVDS is a digital, wired transmission technology for the short range which adopts a differential interface having the characteristics of low voltage and low power consumption.

*2 : HDCP (High-bandwidth Digital Content Protection system) is a copyright protection technology which prevents content from being copied fraudulently. The pathway in a display device such as a display, which sends/receives digital signals from a video playback device such as a computer, is encrypted.

Temperature detection function

- The center display detects temperature using a temperature sensor.
- The center display converts the temperature detected by the temperature sensor to an electrical signal (LVDS signal) and sends it to the CMU.

Note

- If the center display receives a temperature warning pop-up screen display request signal (LVDS signal) from the CMU, it displays the temperature warning pop-up screen on the LCD.

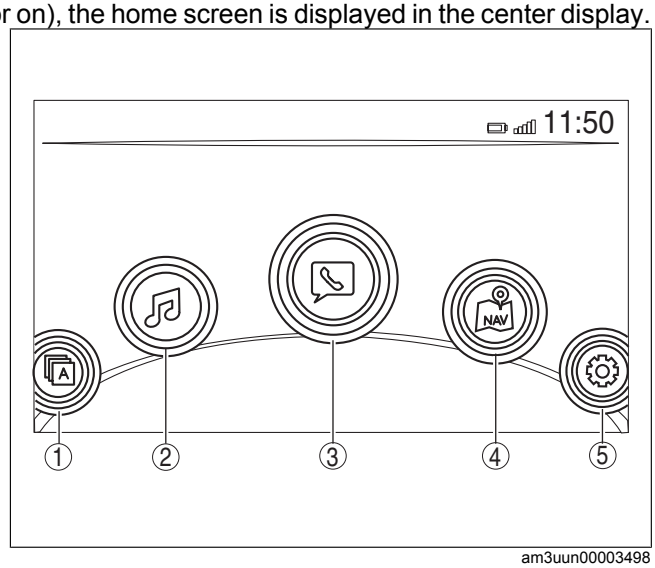
Display information

- The information displayed by the center display is as follows: Refer to the referenced section for details of the displayed information.

Display content	Reference
Home screen	(See Home screen.)
Application screen	(See Applications screen.)
Entertainment screen	(See Entertainment screen.)
Communication screen	(See Communication screen.)
Navigation screen	(See Navigation screen.)
Setting screen	(See Settings screen.)

Home screen

- The home screen displays entertainment system information.
- When the ignition is switched to ACC or ON (engine off or on), the home screen is displayed in the center display.



am3uun00003498

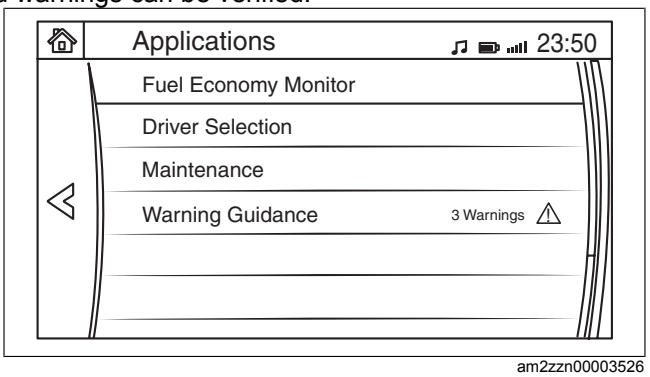
Display area	Button name
1	Applications

2	Entertainment
3	Communication
4	Navigation*3
5	Settings

*3 : With car-navigation system

Applications screen

- Information concerning fuel economy, maintenance, and warnings can be verified.



am2zzn00003526

Display content	Reference
Fuel Economy Monitor	(See Fuel Economy Monitor.)
Maintenance	(See MAINTENANCE MONITOR [WITH CENTER DISPLAY].)
Warning Guidance	(See Warning guidance.)

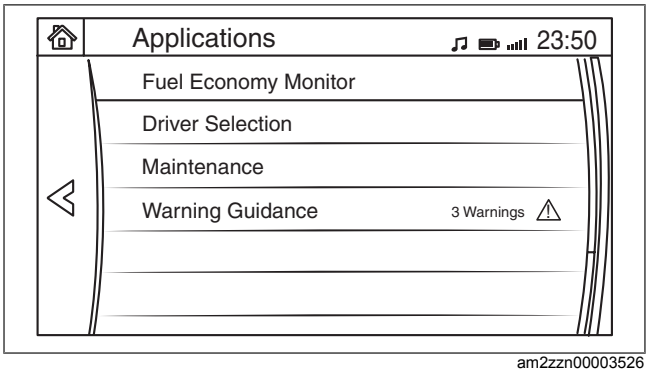
Fuel Economy Monitor

- Displays the fuel economy and i-ELOOP system information.

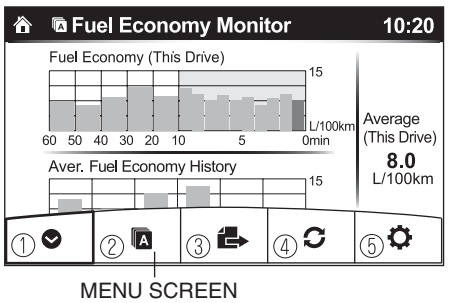
Fuel economy monitor verification procedure

1. Select the “Applications” button on the Home screen.
2. Select [Fuel Economy Monitor] on the Applications screen to display the warning guidance.

- The buttons on the menu screen operate as follows:



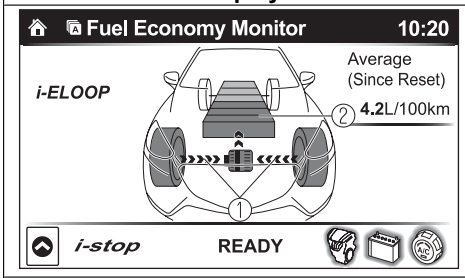
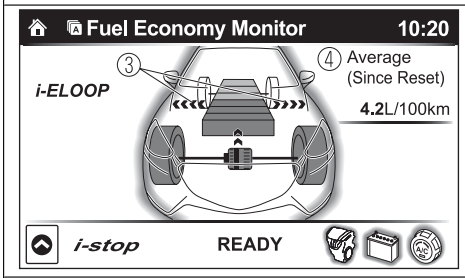
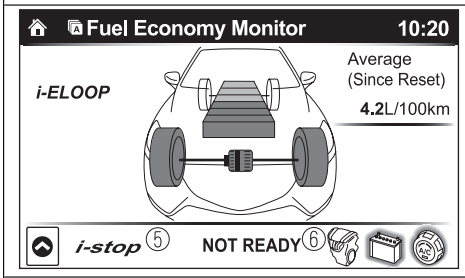
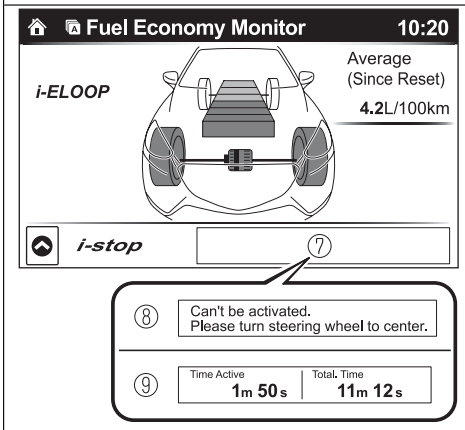
am2zzn00003526

Display		Content
	1	Switches the menu screen to non-display.
	2	Displays the application menu screen.
	3	• Switches the fuel economy monitor display in the following order: — Control Status — Fuel Consumption — Effectiveness
	4	Resets the fuel economy data.
	5	• Displays the following setting screen. — Ending screen on/off switching — On/off switching for function which synchronizes reset fuel economy data to trip meter.

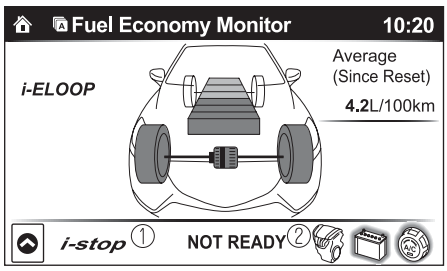
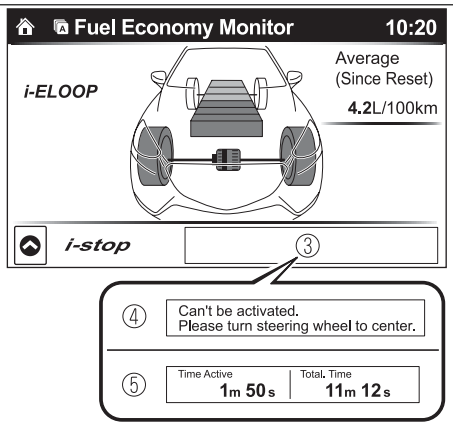
Control Status

- The i-stop operation status and the i-ELOOP power generating status are displayed.

With i-ELOOP system

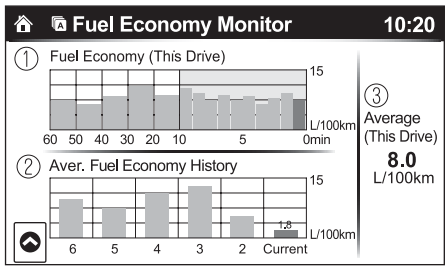
Display		Content	Reference
	1	Displays the level of electricity generated by regenerative braking.	(See i-ELOOP [i-ELOOP].) (See i-stop CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See i-stop CONTROL [SKYACTIV-D 1.5].)
	2	Displays the amount of the electricity stored in the capacitor.	
	3	Displays the status of the electricity stored in the capacitor and being supplied to the electrical devices (whole vehicle in display is illuminated simultaneously).	
	4	Displays the accumulated fuel economy.	
	5	Displays the readiness status as to whether operation of the i-stop function is available.	
	6	Displays the operation readiness status of the i-stop function on the vehicle side (engine, battery, and A/C) using icon colouring. The blue colour indicates that the i-stop function is ready to operate, and the grey colour indicates that it is not ready for operation.	
	7	Displays the i-stop function status while the vehicle is stopped.	
	8	Indicates the operations required by the driver to operate the i-stop function while it is not operating.	
	9	**Time active** • Current i-stop time is displayed i-stop time at which the i-stop is operating. **Total time** • i-stop total time is displayed as the total time at which the i-stop has operated.	

Without i-ELOOP system

Display		Content	Reference
	1	Displays the readiness status as to whether operation of the i-stop function is available.	—
	2	Displays the operation readiness status of the i-stop function on the vehicle side (engine, battery, and A/C) using icon colouring. The blue colour indicates that the i-stop function is ready to operate, and the grey colour indicates that it is not ready for operation.	
	3	Displays the i-stop function status while the vehicle is stopped.	
	4	Indicates the operations required by the driver to operate the i-stop function while it is not operating.	
	5	**Time active** • Current i-stop time is displayed i-stop time at which the i-stop is operating. **Total time** • i-stop total time is displayed as the total time at which the i-stop has operated.	

Fuel Consumption

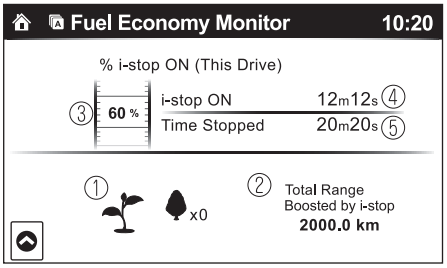
- The information regarding the fuel economy is displayed.

Display		Content	Reference
	1	• Displays the fuel economy for the past 60 minutes. — Displays the fuel economy every minute for the past 1 to 10 minutes. — Displays the fuel economy every 10 minutes for the past 10 to 60 minutes.	—
	2	Displays the average fuel economy over the past 5 resets and after the current reset.	
	3	• Displays the average fuel economy from when the ignition is switched ON (engine on) to the current time. • The average fuel economy is updated every 60 s. • If there is no average fuel economy data, displays "— —".	

Effectiveness

- The actual performance of the energy efficiency is displayed.

Vehicles with i-ELOOP system and i-stop function

Display		Content	Reference
	1	The leaf graphic grows according to the amount of CO2 emissions reduced by the system effects. The cumulative total amount of leaf growth is indicated in terms of the number of tree graphics.	—
	2	Displays the total driving distance which could be extended by the operation of the i-stop function.	
	3	Displays the percentage of time that the vehicle was stopped by the i-stop function operation out of the total amount of time that the vehicle was stopped.	
	4	Displays the amount of time that the i-stop function has operated.	
	5	Displays the total amount of time that the vehicle was stopped.	

Warning guidance

- When the center display receives a warning guidance display signal (LVDS signal) from the CMU, it displays the warning guidance screen.

Note

- Display method of the warning guidance varies depending on the emergency level.

Pop-up display

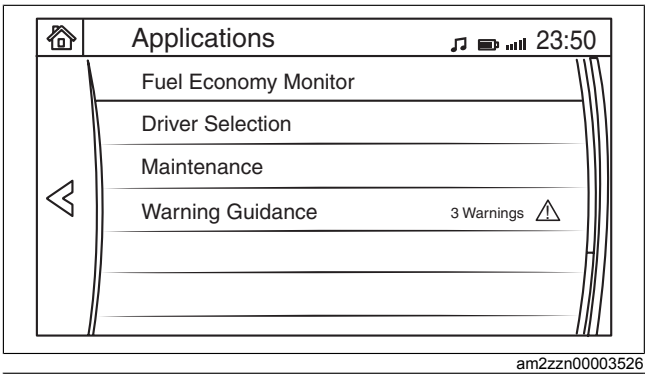
— Warning message of a higher emergency level is displayed on the pop-up screen at the time the warning occurred regardless of the display content of the center display. When “OK” on the pop-up screen is selected, it switches to the previous screen before the pop-up display. The pop-up display warning can also be verified in the warning guidance list.

List display

— Warning messages displayed by methods other than pop-up display can be verified in the warning guidance screen.

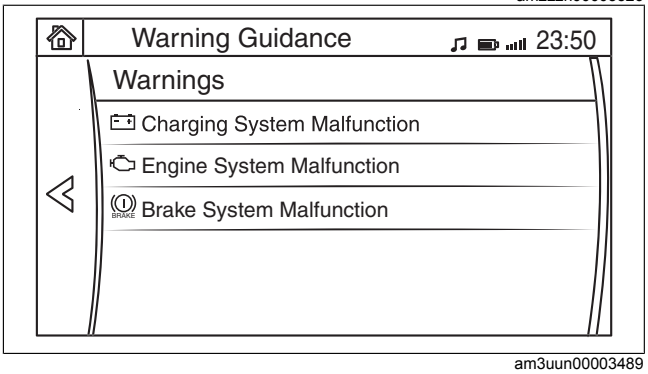
Warning guidance verification procedure

1. Select the [Applications] button on the Home screen.
2. Select [Warning Guidance] on the Applications screen to display the warning guidance.



am2zzn00003526

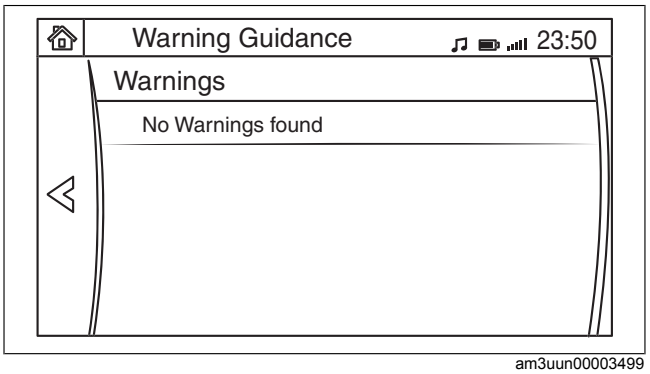
3. Select a desired warning from the warning guidance screen.



am3uun00003489

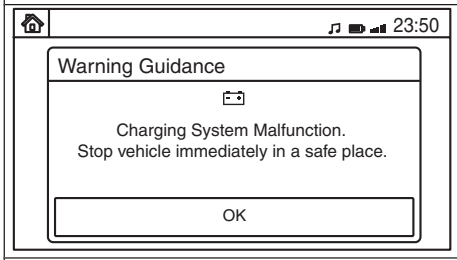
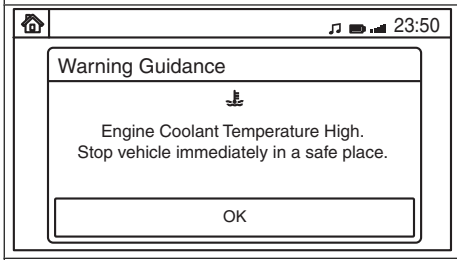
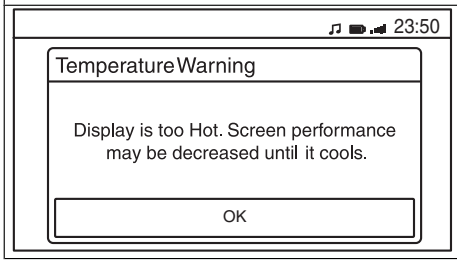
Note

- If there is no warning, No Warnings found is displayed on the warning guidance screen.



am3uun00003499

Pop-up display

Screen display			Send module	Reference/function
Pop-up screen	List display			
	Title	Message		
	Charging System Malfunction	There is an error in the charging system Stop car immediately in a safe place, and contact a Mazda dealer.	PCM	(See CHARGING SYSTEM WARNING LIGHT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CHARGING SYSTEM WARNING LIGHT [SKYACTIV-D 1.5].)
	Engine Coolant Temperature High (On)	Engine Coolant Temperature High. Stop vehicle immediately in a safe place.	Instrument cluster	(See LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT (BLUE)/ HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT (RED).)
	—*4	—*4	CMU/ center display	—

*4 : Because the temperature warning is not displayed in the list, it cannot be verified using the list.

List display

Screen display			Send module	Reference/function
Warning light	Title	Message		
	Low Engine Oil Pressure	Low engine oil pressure. Stop the vehicle immediately in a safe place, turn off the engine, and inspect the engine oil level. Add oil if necessary. Have the vehicle checked by a dealer immediately if this indicator turns on after adding oil or the engine oil level is OK.	PCM	(See ENGINE OIL WARNING LIGHT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ENGINE OIL WARNING LIGHT [SKYACTIV-D 1.5].)
	i-ELOOP System Malfunction	i-ELOOP system malfunction. Have the vehicle checked by a dealer if this indicator stays on.	PCM (i-ELOOP)	(See MASTER WARNING LIGHT.)

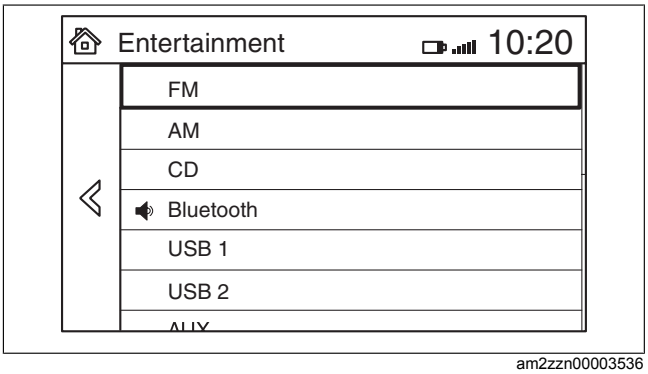
Screen display			Send module	Reference/function
Warning light	Title	Message		
	Low Brake Fluid	Brake fluid level is low. Stop the vehicle immediately in a safe place and have the vehicle checked by a dealer.	Front body control module (FBCM)	(See BRAKE SYSTEM/ ABS WARNING LIGHT.)
	Brake System Malfunction	Electronically-controlled brake system malfunction. Stop the vehicle immediately in a safe place and have the vehicle checked by a dealer.	DSC HU/CM	
	Low Tire Air Pressure (On)	Low tire air pressure in one or more tires. Check the air pressure of the tires and adjust to the correct air pressure. Press the TPMS SET switch after adjusting the tire pressure until you hear a beep.	DSC HU/CM	(See TIRE PRESSURE MONITORING SYSTEM WARNING LIGHT.)
	Engine Coolant Temperature High	Engine is starting to run hot. Drive slowly or stop the vehicle until the indicator stops flashing. If it keeps flashing, even after stopping the vehicle, take proper actions. Refer to the Owner's Manual.	Instrument cluster	(See LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT (BLUE)/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT (RED).)
DPF	PM Accumulation in DPF	The particulate matter (PM) collected in the diesel particulate filter (DPF) has exceeded the regulation value. Drive for about 10 to 15 minutes with the engine fully warmed-up (blue temp light off) at a speed of 10 mph (15km/h) or more to eliminate the PM. Have the vehicle checked by a dealer if this indicator stays on.	PCM	(See DIESEL PARTICULATE FILTER INDICATOR LIGHT [SKYACTIV-D 1.5].)
DPF	DPF Error	Diesel particulate filter (DPF) malfunction. Have the vehicle checked by a dealer.	PCM	
	Low Fuel	Fuel tank level is low. Refuel the vehicle soon.	Instrument cluster	(See LOW FUEL WARNING LIGHT.)
	Power Steering System Malfunction	Electric power steering malfunction. Stop the vehicle in a safe place and turn off the engine. Have the vehicle checked by a dealer if this indicator stays on.	EPS control module	(See POWER STEERING MALFUNCTION INDICATOR LIGHT.)
	Engine System Malfunction	Engine system malfunction. Have the vehicle checked by a dealer immediately.	PCM	(See CHECK ENGINE LIGHT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CHECK ENGINE LIGHT [SKYACTIV-D 1.5].)
	Airbag System/ Pretensioner System Malfunction	Airbag/seatbelt malfunction. Have the vehicle checked by a dealer immediately	SAS control module	(See AIR BAG/SEAT BELT PRE-TENSIONER SYSTEM WARNING LIGHT [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See AIR BAG/SEAT BELT PRE-TENSIONER SYSTEM WARNING LIGHT [STANDARD DEPLOYMENT CONTROL SYSTEM].)
	ABS Malfunction	Anti-lock brake system (ABS) malfunction. Have the vehicle checked by a dealer immediately.	DSC HU/CM	(See BRAKE SYSTEM/ ABS WARNING LIGHT.)
	DSC Malfunction	Dynamic stability control (DSC) or traction control system (TCS) malfunction. Have the vehicle checked by a dealer immediately.	DSC HU/CM	(See TCS/DSC INDICATOR LIGHT.)

Screen display			Send module	Reference/function
Warning light	Title	Message		
	BSM Malfunction	Blind spot monitoring (BSM)/Rear cross traffic alert (RCTA) malfunction. Check for abnormalities in the rear bumper surface and clean if dirty or covered with snow. Have the vehicle checked by a dealer if this indicator stays on.	BSM control module	(See BLIND SPOT MONITORING (BSM) SYSTEM.)
	TPMS System Malfunction	Tire pressure monitoring system (TPMS) malfunction. Have the vehicle checked by a dealer.	DSC HU/CM	(See TIRE PRESSURE MONITORING SYSTEM WARNING LIGHT.)
i-stop	i-Stop System Malfunction	i-stop system mulfunction. Have the vehicle checked by a dealer.	PCM	(See i-stop INDICATOR LIGHT (GREEN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See i-stop WARNING LIGHT (AMBER) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See i-stop INDICATOR LIGHT (GREEN) [SKYACTIV-D 1.5].) (See i-stop WARNING LIGHT (AMBER) [SKYACTIV-D 1.5].)
	Key Battery Low	Key battery is low. Replace the key battery immediately.	Start stop unit	(See KEY WARNING LIGHT (RED)/KEY INDICATOR LIGHT (GREEN).)
	Keyless Entry Malfunction	Keyless entry system malfunction. Have the vehicle checked by a dealer.	Start stop unit	
AT	Automatic Transmission Malfunction	Automatic transmission malfunction. Damage to the vehicle may occur if you continue to drive. Have the vehicle checked by a dealer immediately.	TCM	(See AUTOMATIC TRANSAXLE WARNING LIGHT [CW6A-EL].) (See AUTOMATIC TRANSAXLE WARNING LIGHT [EW6A-EL].)
	Battery Management System Malfunction	Battery management system malfunction. Have the vehicle checked by a dealer.	PCM	(See MASTER WARNING LIGHT.)
	SCBS Malfunction	Smart city brake support (SCBS) malfunction. Have the vehicle checked by a dealer immediately.	Laser sensor	
	SCBS Malfunction	Smart city brake support (SCBS) malfunction. Clean the front windshield. Have the vehicle checked by a dealer if this indicator stays on.	Laser sensor	
	SCBS Unavailable	Smart city brake support (SCBS) currently unavailable.	Laser sensor	
	LDWS Malfunction	Lane departure warning system (LDWS) malfunction. Have the vehicle checked by a dealer.	Forward sensing camera (FSC)	(See LANE DEPARTURE WARNING SYSTEM (LDWS).)
	HBC Malfunction	High beam control (HBC) malfunction. Have the vehicle checked by a dealer.	Forward sensing camera (FSC)	(See HBC INDICATOR LIGHT (GREEN)/HBC WARNING LIGHT (AMBER).)

Screen display			Send module	Reference/function
Warning light	Title	Message		
⚠	HBC/ LDWS Camera Malfunction	High beam control (HBC)/Lane departure warning system (LDWS) malfunction. Clean the windshield. Have the vehicle checked by a dealer if this indicator stays on.	Forward sensing camera (FSC)	(See MASTER WARNING LIGHT.)
⚠	HBC/ LDWS System Malfunction	High beam control (HBC)/Lane departure warning system (LDWS) malfunction. Have the vehicle checked by a dealer if this indicator stays on.	Forward sensing camera (FSC)	
⚠	HBC/ LDWS System Malfunction	High beam control (HBC)/Lane departure warning system (LDWS) malfunction. Defrost the windshield. Have the vehicle checked by a dealer if this indicator stays on.	Forward sensing camera (FSC)	
⚠	Vehicle System Malfunction	Vehicle system malfunction. Have the vehicle checked by a dealer.	PCM	
	Low Washer Fluid	Washer fluid level is low. Refill the washer fluid.	Front body control module (FBCM)	(See WASHER FLUID-LEVEL SENSOR.)
	Engine Coolant Temperature Low	Engine coolant is cold and no warm air is available for heater/ defroster. Vehicle can be driven with the indicator on or off. Have the vehicle checked by a dealer if this indicator stays on.	Instrument cluster	(See LOW ENGINE COOLANT TEMPERATURE INDICATOR LIGHT (BLUE)/HIGH ENGINE COOLANT TEMPERATURE WARNING LIGHT (RED).)
	Glow plugs preheating	Glow plugs are preheating. Have the vehicle checked by a dealer if this indicator stays on.	PCM	(See GLOW INDICATOR LIGHT [SKYACTIV-D 1.5].)

Entertainment screen

- When the entertainment button on the home screen is selected, the entertainment screen is displayed. The following audio functions in the table below are available.



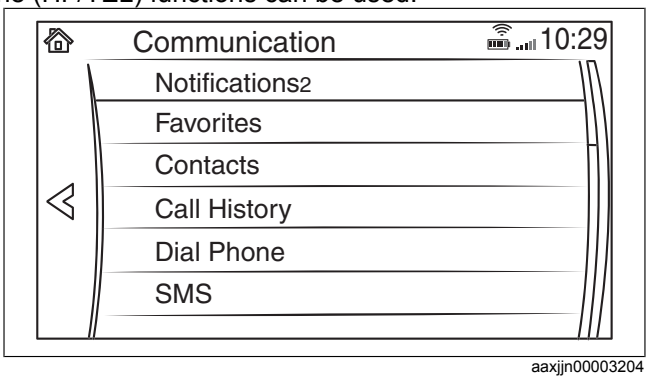
am2zzn00003536

Display content	Function
FM	Switches to FM radio screen.
AM	Switches to AM radio screen.

Display content	Function
Stitcher™	Switches to Stitcher. **Note** • Stitcher™ is not displayed in gray even if the connected mobile device is not compatible with Stitcher™.
Aha™ Radio	Switches to Aha™ radio screen. **Note** • Aha™ radio is not displayed in gray even if the connected mobile device is not compatible with Aha™ radio.
Bluetooth®<Device Name>	Switches to Bluetooth® audio screen. **Note** • If available, the connecting device name is displayed. When Bluetooth® is not available, such as no device is connected, only Bluetooth® is displayed.
USB1<Device Name>	Switches to USB audio screen. **Note** • If available, the connecting device name is displayed. When USB1 is not available, such as no device is connected, only USB1 is displayed.
USB2<Device Name>	Switches to USB audio screen. **Note** • If available, the connecting device name is displayed. When USB2 is not available, such as no device is connected, only USB2 is displayed.
CD	Switches to CD screen.
AUX	Switches to AUX screen.
DVD	Switches to DVD screen.

Note
- Items requiring media, such as CDs or AUX, are displayed in gray and not available if connection of the media cannot be detected.

Communication screen
- SMS (Short Message Service), and Hands-free telephone (HF/TEL) functions can be used.



aaxjjn00003204

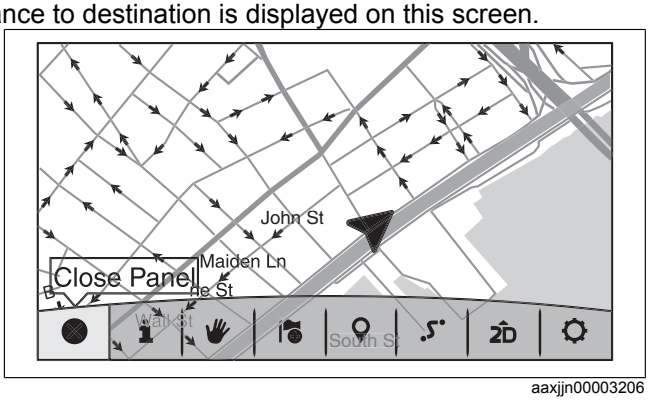
Display content	Function
Active Call	Switches to the incoming call screen. (when receiving an incoming call)
Notifications	Switches to the notification screen. **Note** • When there is no notification, displayed in gray and not available.
Favorites	Switches to the favorite screen.
Contacts	Displays the phone book.
Call History	Displays the call history.
Dial Phoned	Switches to the dial screen for phone number input.
SMS	Switches to the SMS screen.
Setting	Switches ON/OFF of the SMS notification function.

Note

- If Bluetooth® communication is not established between a mobile device such as a Bluetooth®-enabled smart phone, the functions dependent on the mobile device are displayed in gray and they are not available.
- IF there any unread notification or SMS, the number of unread messages is displayed beside each item name. The number of missed call is displayed next to “Call History”.

Navigation screen

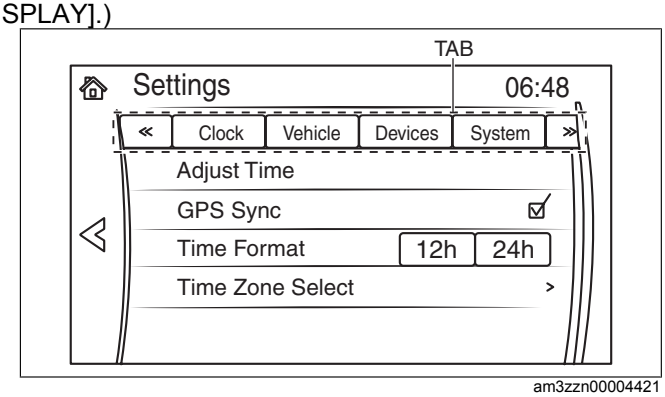
- When the navigation button on the home screen is selected, the navigation screen is displayed. Navigation information such as the current position and route guidance to destination is displayed on this screen.



aaxjjn00003206

Settings screen

- The following settings can be changed. For details, refer to the ENTERTAINMENT SYSTEM setting function. (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)

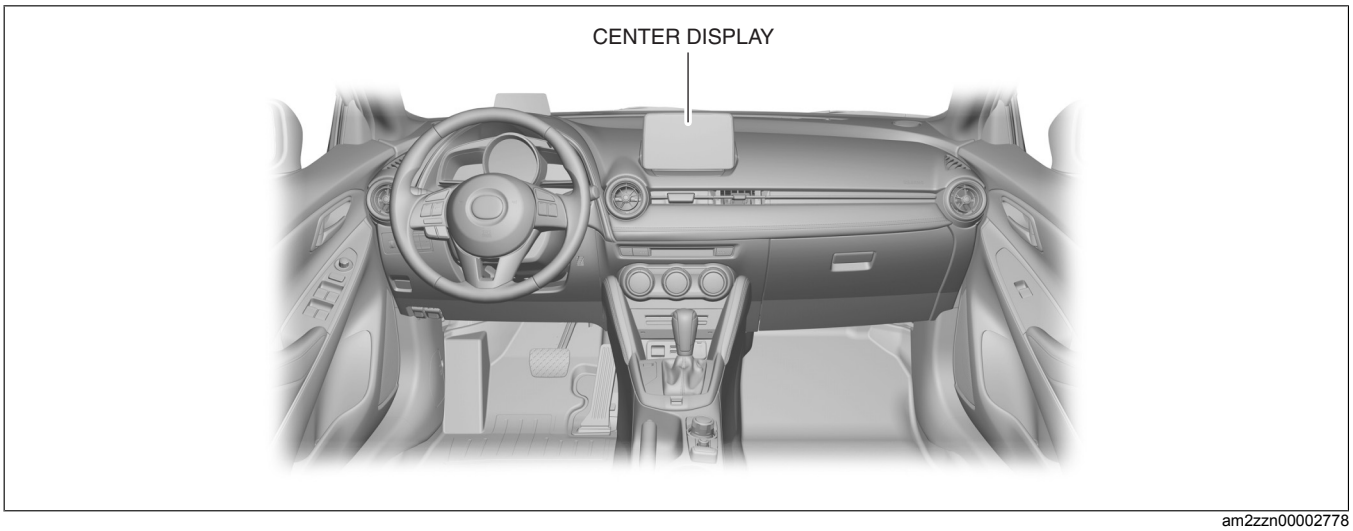


am3zzn00004421

Tab	Reference	
AD-Disp	(See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)	Active driving display setting
Display		Display setting
Safety		Personalization setting function (vehicle setting)
Sound		Sound setting
Clock		Clock setting
Vehicle		Personalization setting function (vehicle setting)
Devices		Bluetooth® pairing
System		System setting

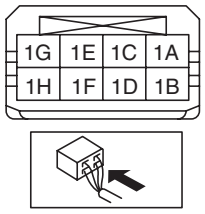
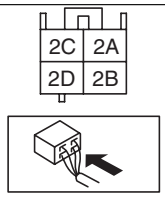
Construction

- The center display is installed in the center of the dashboard.
- The center display has a built-in temperature sensor.



am2zzn00002778

Input/output signal table

Terminal		Signal
	1A	—
	1B	Ground
	1C	—
	1D	—
	1E	—
	1F	—
	1G	ACC
	1H	B+
	2A	—
	2B	LVDS (+)
	2C	—
	2D	LVDS (-)

Fail-safe

- Not applicable